P1. Assignment 1.3 Admino Gradescope

Linux Administration (admino)

The objective of this lab is to practice security related linux administration functions. These functions will provide different types of **security-related information**, which may be extremely useful for a Linux System Administrator. In Linux systems, there is a huge amount of system information available that sometimes makes harder to find specific information related to a cyber incident. For that reason, it is very important to know how information and data is stored and how to appropriately filter that massive amount of data to extract only what's needed. This capacity is essential to obtain the required piece of information relevant to your cybersecurity objectives.

Linux operating systems provide an impressive amount of tools to help you to filter information from files and command outputs. This assignment will help you learn how they work and how to use them. These are some of the most relevant command-line utilities.

- grep - Matching a pattern in file or output
- awk - For pattern scanning and processing language
- sed - For filtering and transforming text
- uniq - Report or omit repeated lines
- sort - Sort lines of text files

If any of these tools are not available for you in your Linux system, you are free to install them (e.g., sudo apt-get install uniq -y)

At the same time, this assignment will also provide you the opportunity to practice bash scripting*conditions*, *arguments*, *variables*, *loops*, ... so you can automate tasks for systems to self-execute them.

Scenario

In this assignment you will be connecting to your individual Linux system hosted in the cloud. Each of you has been assigned an individual port for connecting.

To connect you will be using SSH

- user: student
- pass:15Champions

Here is an example of the connection (remember that you need to use the port assigned to you for connecting):

$ssh student@cy5001p1.mscy.org -p 220XX

Your assigned port can be found here:

Deliverables

Your deliverables for this assignment consist of:

1. A text file named user#.adminos26.txt submitted through **Gradescope (Project 1.3 assigned PORTS)**. This file needs to have one string per line to answer each task **in the exact order** of the tasks.

```
>$cat user#.adminos26.txt
cy5001{RBINio4839q39jiorrptuwuwptupotw==}
Jun 06 17:53:01 cy5001server docker[22876]: Failed password for backup
cy5001{RSYSaXbAOiH9foeLlPWXv1vSoQ==}
cy5001{RACC1418QmnVDe9jvuQafQRs0w==}
cy5001{RENVXbAOiH9foeLlPWXv1vSoQ==}
15119
0012345678:4e5be14c1f728e9d26c0b38c35b14d7fc7a313f7
```

2. An asciinema .cast file named "***user#***_T7.cast", For user**#** . Your assigned port will determine that number #, so if your port is 220**87**, your user will be **user87**. If your port is 22008 your username will be **user8.** Your port to determine **your user# is here.** Your cast file needs to record the completion of the Task 7. It does not need to be a clean recording, but it needs to achieve the goal of the task during the recording.

The GRADE provided in GradeScope is not final, there will be manual review of all assignments before providing the final grade

Tasks to complete

In order to complete your assignment you need to provide the proper answers to the following task.

1. Malware techniques. User's privileges, file's permissions and hidden files

- Malware often hide information using different tools regularly provided by Linux operating system.
- In your system there is a hidden file named data.bin. First, you need to find it in your system and then use Linux administration tools to identify a string in it containing your

individual flag. The flag looks like this:

- cy5001{RBINZiV/MiFm9jQyOwyKmHtnSQ==}

2. Finding tracks of incidents in logs

- Syslog is a Linux service intended to store significant system events. Your individual system has stored pre-generated logs events in the default folder.
- There are ~1000 events recorded, use Linux tools to filter all the events and find the one that is duplicated. It looks like this:
  - Jun 04 09:32:23 cy5001server ftp[15236]: session opened for user operator
- That repeated event will be your answer for this task.
- Make sure to check ALL the syslog files.

3. Finding specific record in syslog events

- Syslog is a Linux service intended to store significant system events. Your individual system has stored pre-generated logs events in the default folder.
- There are ~1000 events recorded, use Linux tools to filter and find a flag included in the description of one particular event.
- That flag is the answer to this task and looks like this:
  - cy5001{RSYSaXbAOiH9foeLlPWXv1vSoQ==}
- Make sure to check ALL the syslog files.

4. Access control through ownership and permission.

- In your system there is a file located in a random folder, which has zero permissions for the file owner and others, and only read permissions for the group owner.
- You need to locate and identify that file first. Then, using your student user, include your user into that group which owns the file and finally get your flag, which looks like this:
  - cy5001{RACCrdbAOiH9foeLlPWXv1vSoQ==}
  - Remember that group membership changes may require a re-login to take effect.

5. Environment Variables also are input to the system

- In your system, there is an environment variable that contains an individual value. The value is a flag that you need to obtain and provide it as the answer for this part of the assignment.
- That flag is the answer to this task and looks like this:
    - cy5001{RENVXbAOiH9foeLlPWXv1vSoQ==}

6. Reducing attack surface of an unmanaged HTTP service

- You need to check the list of ports that are open in your system. Of course one of them is the SSH port you connected to, but there are another port which is running an HTTP server. You need to identify the correct port and provide that number as the part of your answer to this task.
- The answer to this task looks like this:
    - 1234

7. Authentication based on public key

- We have stored a private key file user into the student home directory. This key is being created with an asymmetric cryptography algorithm, which we will see in our following module. For this task, you just need to know that you are providing this file as a probe of authentication for an SSH connection. For this task your username will be unique (not student) , but user**#** . Your assigned port will determine that number #, so if your port is 220**87**, your user will be **user87**. If your port is 22008 your username will be **user8.**
- **Bring that private key from your remote VM to your laptop**, and from there use it to connect with **SSH to the following server (cy5001p1.mscy.org at port 22100).** Remember that as part of the deliverables you need to provide an asciinema recording of the successful access to the system at port 22100. The recording needs to be named "***user#***_T7.cast", where is your user# is the user name you used to connect to the system.

Once you are in the remote system, you need to find a script named cookie and execute it providing your NUID as argument

>$ cookie 00123456789
00123456789:fefb4f4847337a1bc28f0bf5e2bbad1ce4c14756

Finding this cookie and executing it is the objective of the task.

Grading:

Below is the rubric for this assignment:

Concept	**Points**
Successful completion of task 1-7 (1.2 points each)	8.4 points
Create and submit a proper asciinema recording of task 7	1.6 points